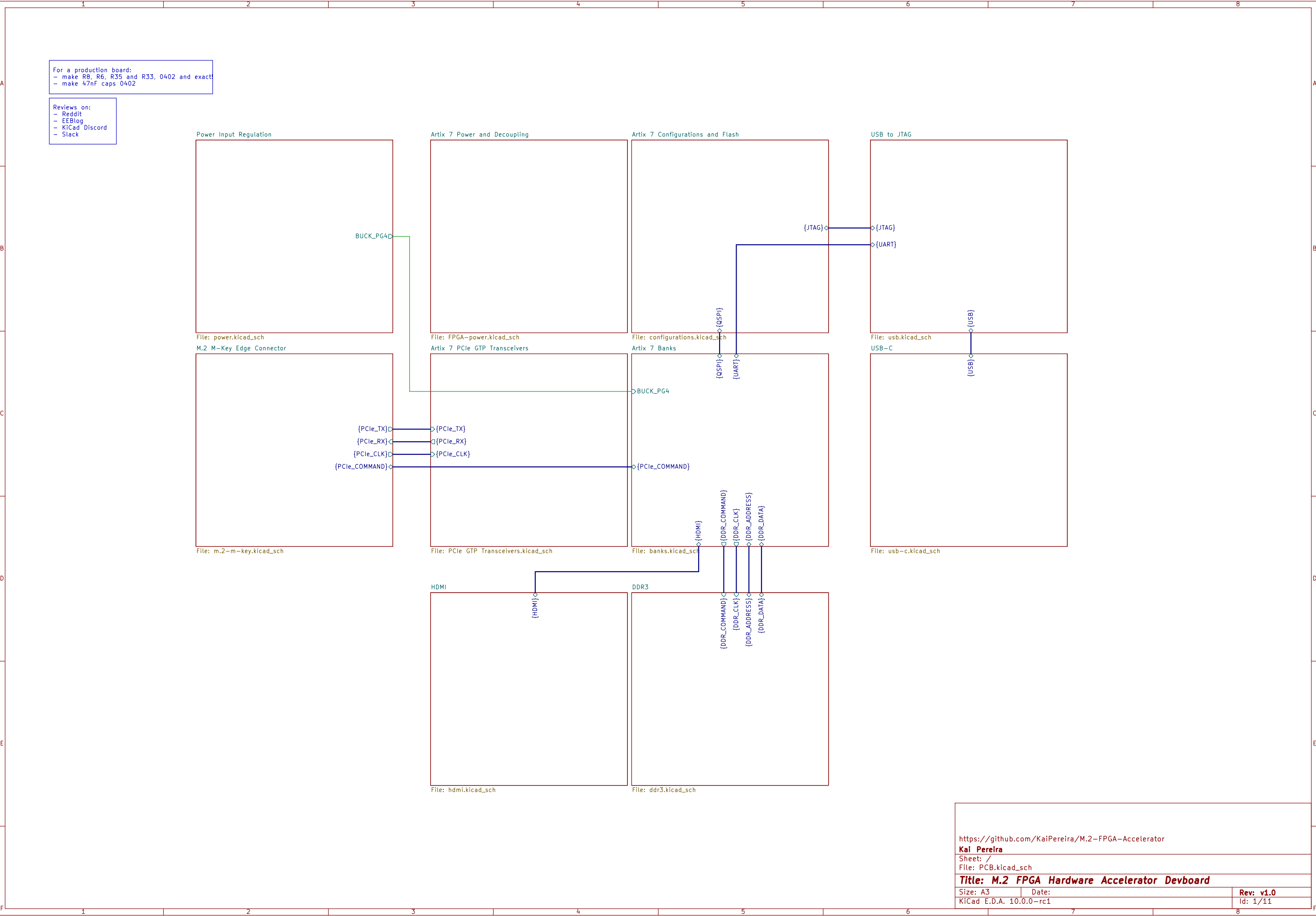
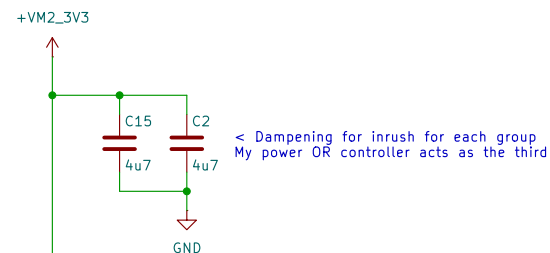




I inverted the bus symbol, to create an edge connector symbol
DOUBLE CHECK the transmit/receive are the right orientation!!
Pg 150, <https://picture.iczhiku.com/resource/eetop/sHksKPigIjgRbbx.pdf>

All the NC's are SATA only features!

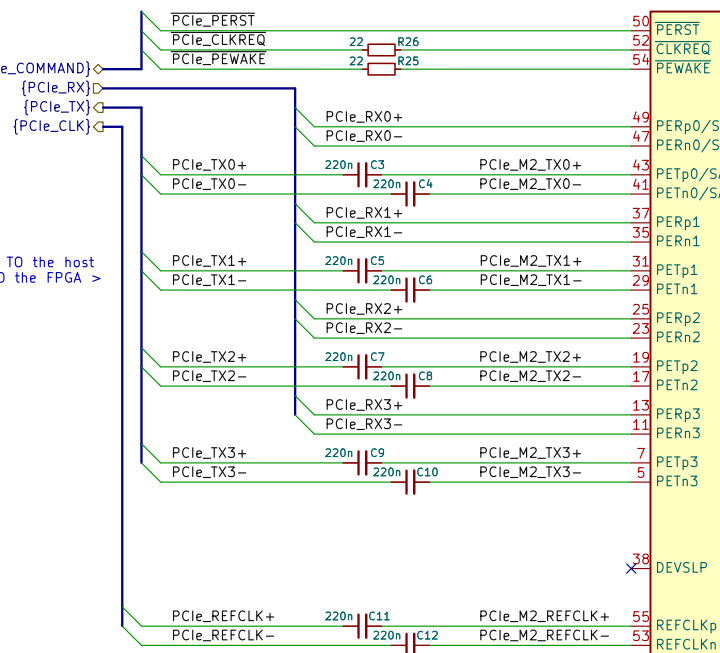
AC Coupling should be near the edge connector to reduce
impedance discontinuities! 220nF for backwards compatibility!



J1
Bus_M.2_Edge_M

Dampening for long, slow traces > {PCle_COMMAND}

Transmitting (TX) FROM the FPGA TO the host
Receiving (RX) FROM the host TO the FPGA >



Sheet: /M.2 M-Key Edge Connector/
File: m.2-m-key.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.0-rc1

Rev:

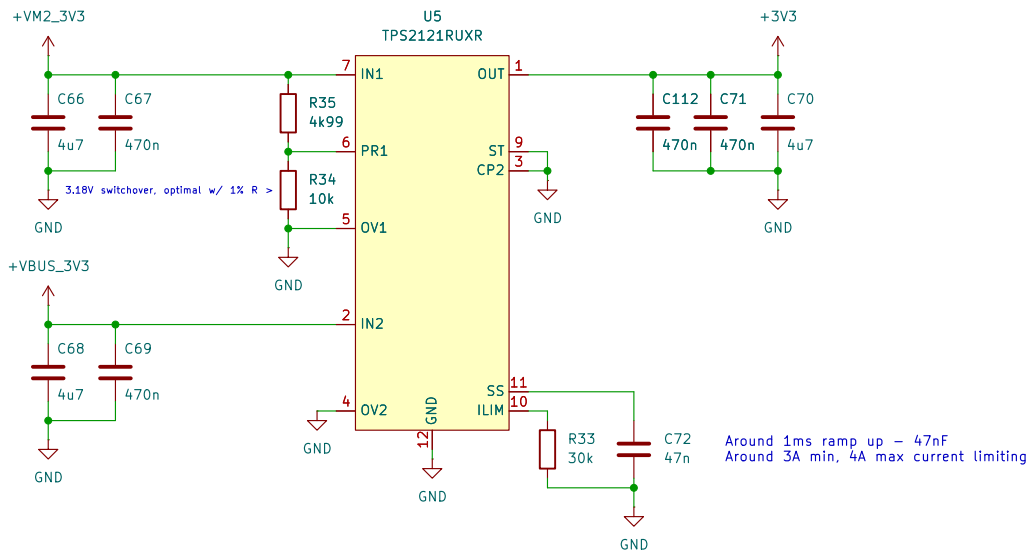
Id: 2/11

Simulate beads in LTSpice!

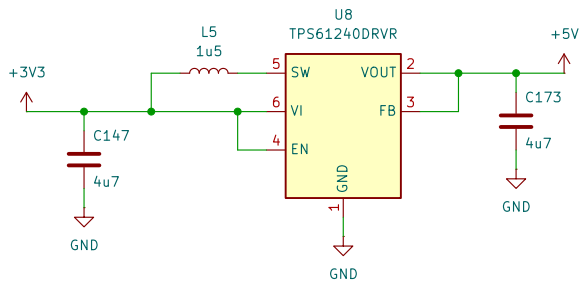
+VM2_3V3 is the M.2 3.3V power source
+VBUS_3V3 is the USB power source, bucked down

This circuit ensures that USB doesn't backfeed into the M.2 connector, frying the motherboard, and also prioritizes reliable M.2 connector power if both are plugged in!

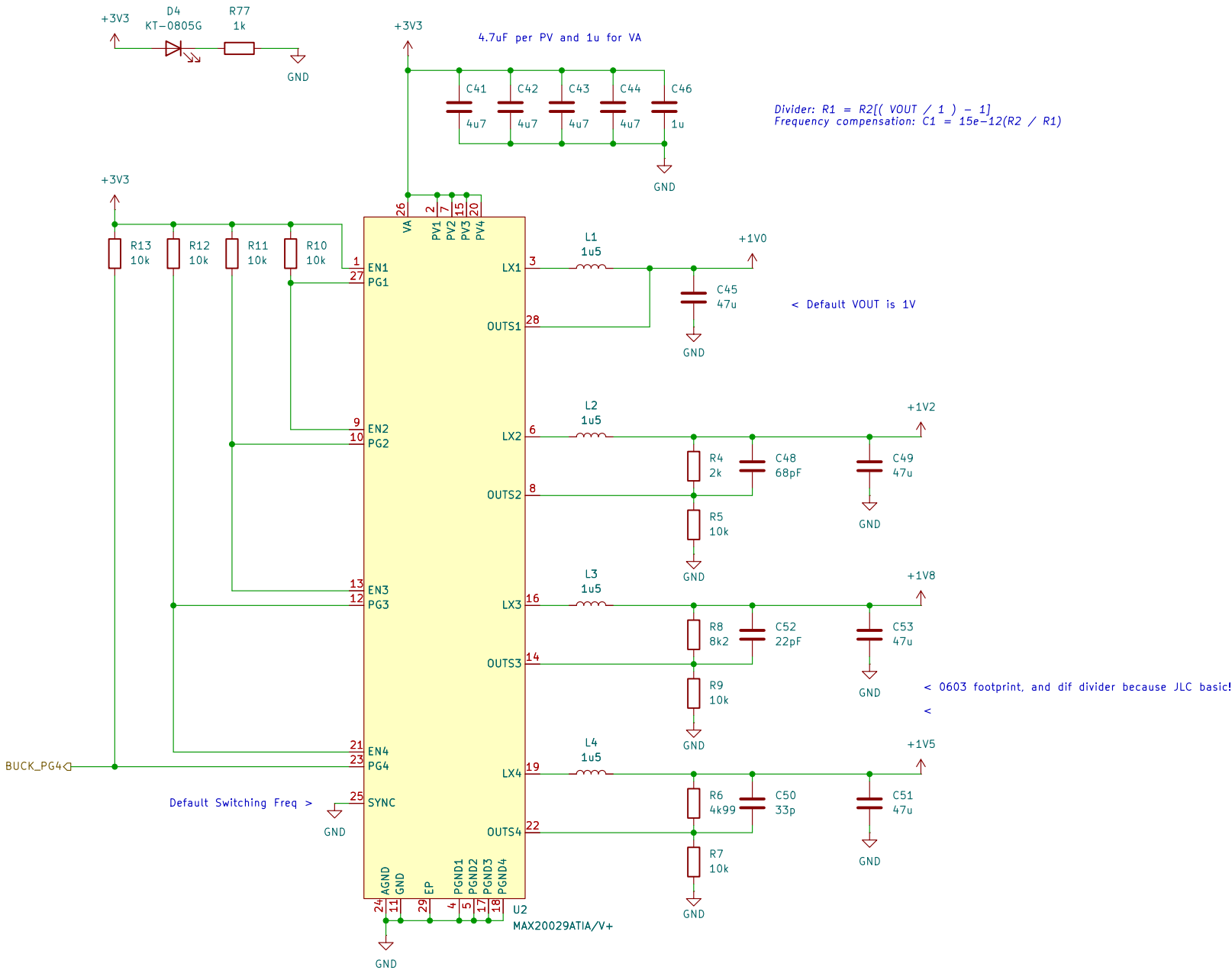
Switchover is at 3.18V using PR1 voltage divider which is really optimal with 1% resistors because m.2 connector minimum voltage is 3.135V

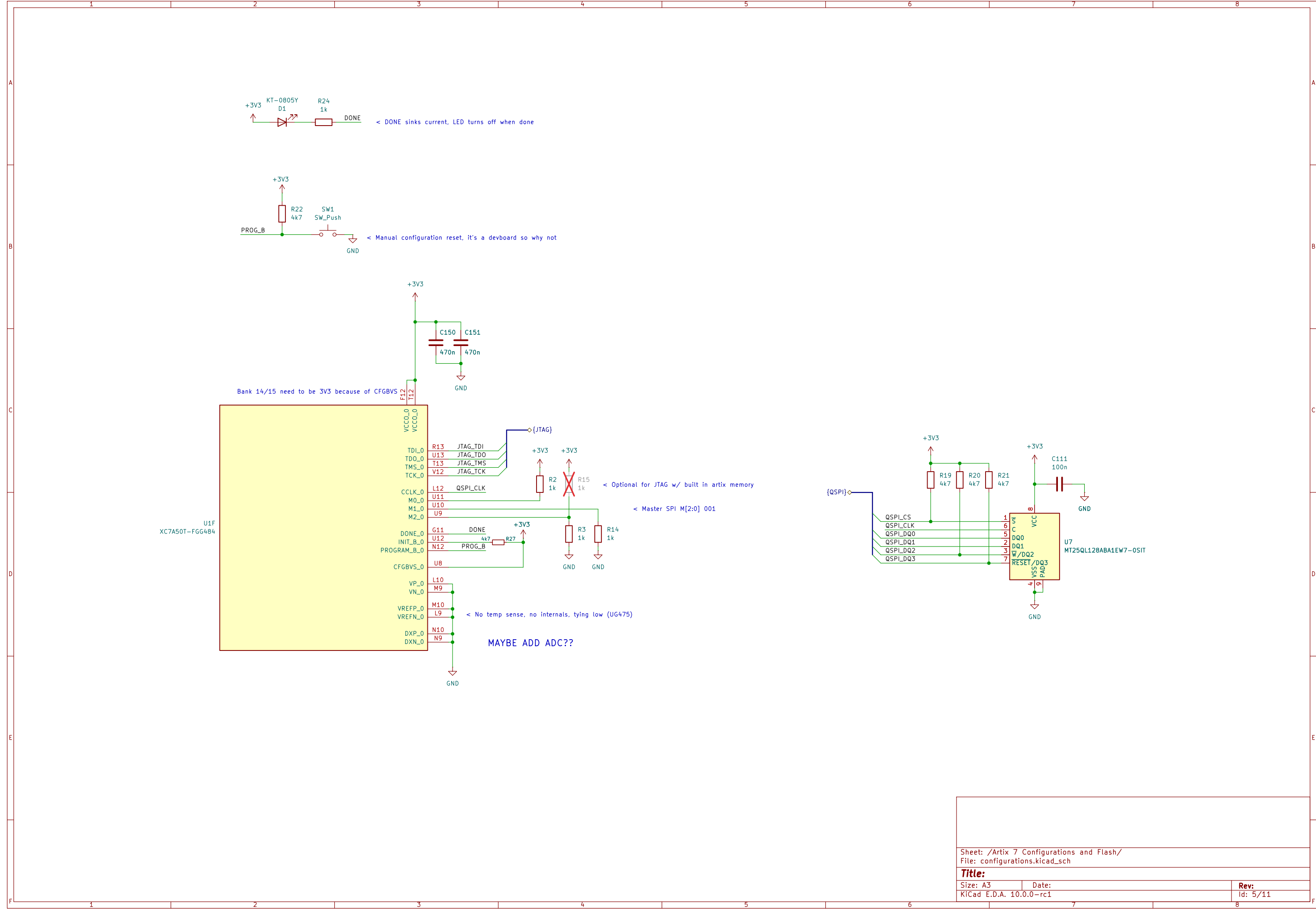


A small 5V boost converter for HDMI and other future peripherals!



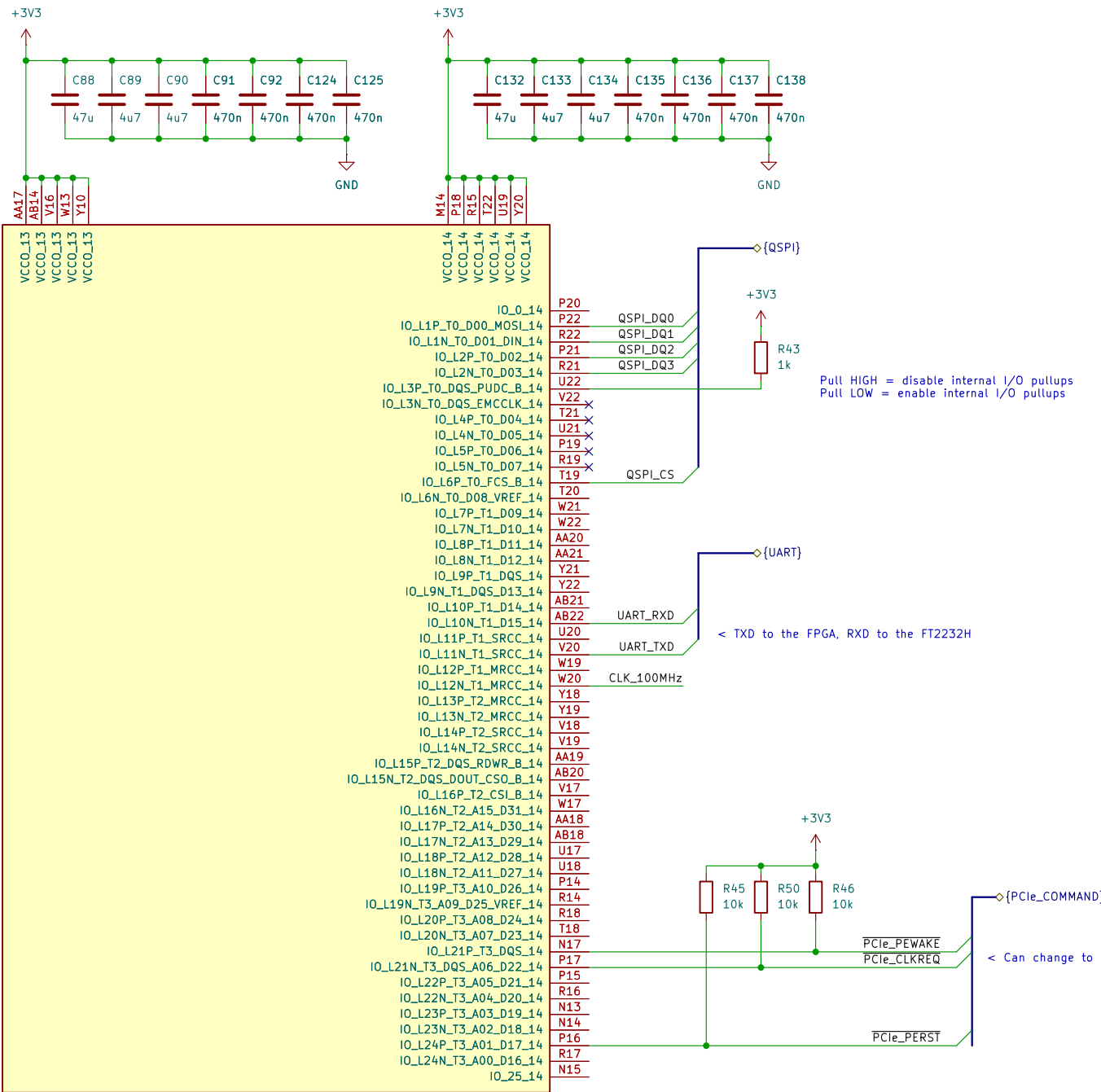
SEQUENCE VCC -> 1V0 -> 1V2 -> 1V8 -> 1V5



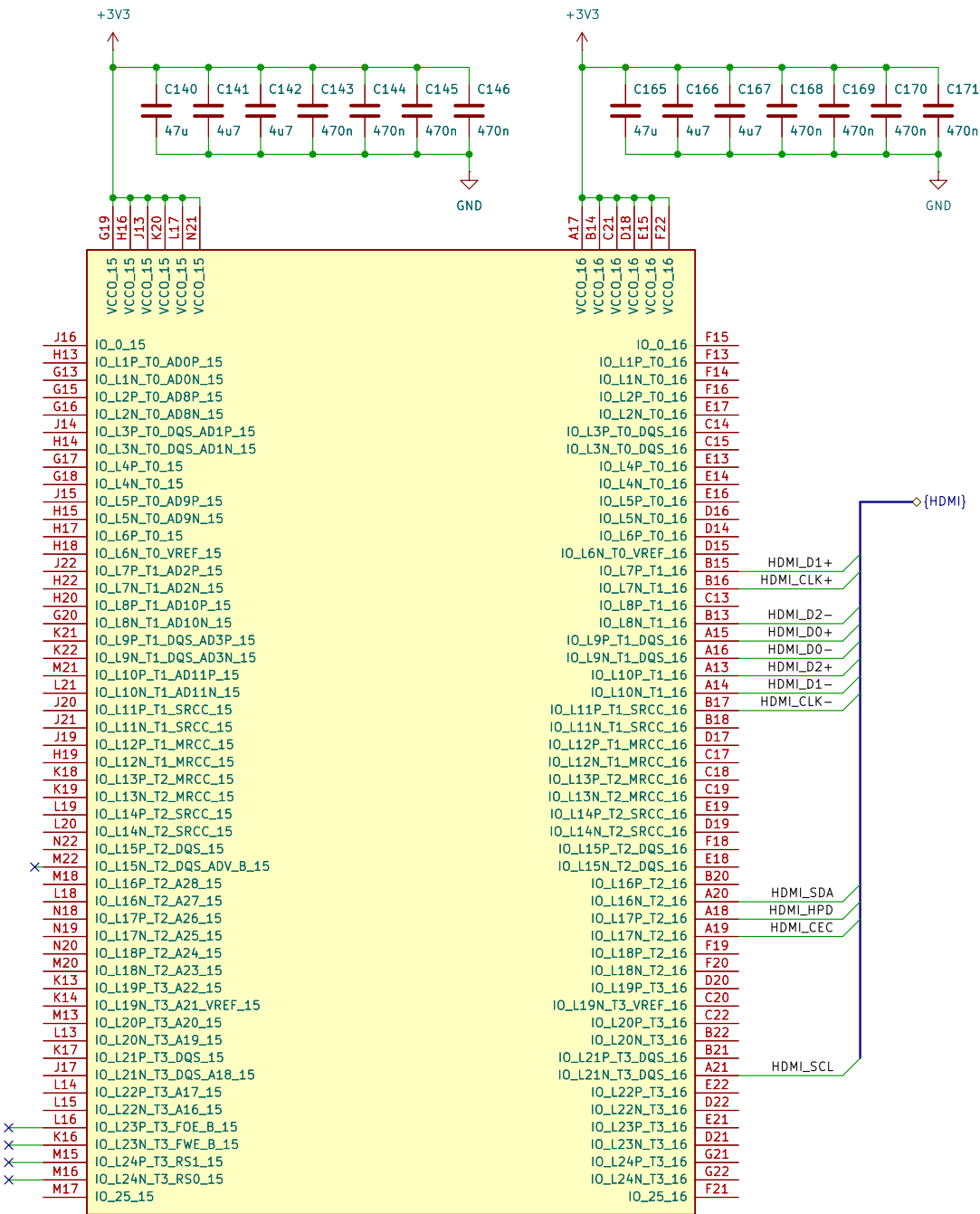


Most of the NC's are usable as I/O's too,
even if they're configurations :)

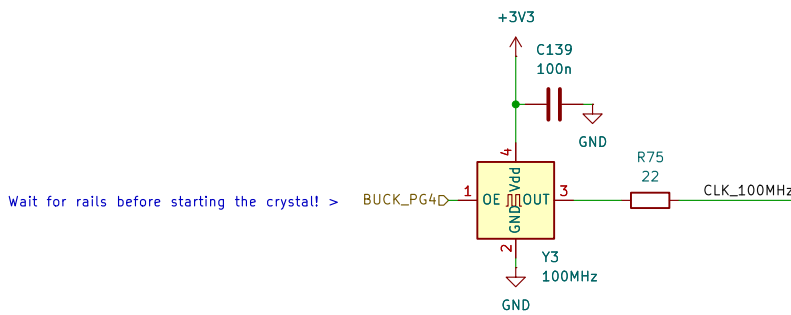
47u + 2x 4u7 + 4x 470n capacitors per bank!

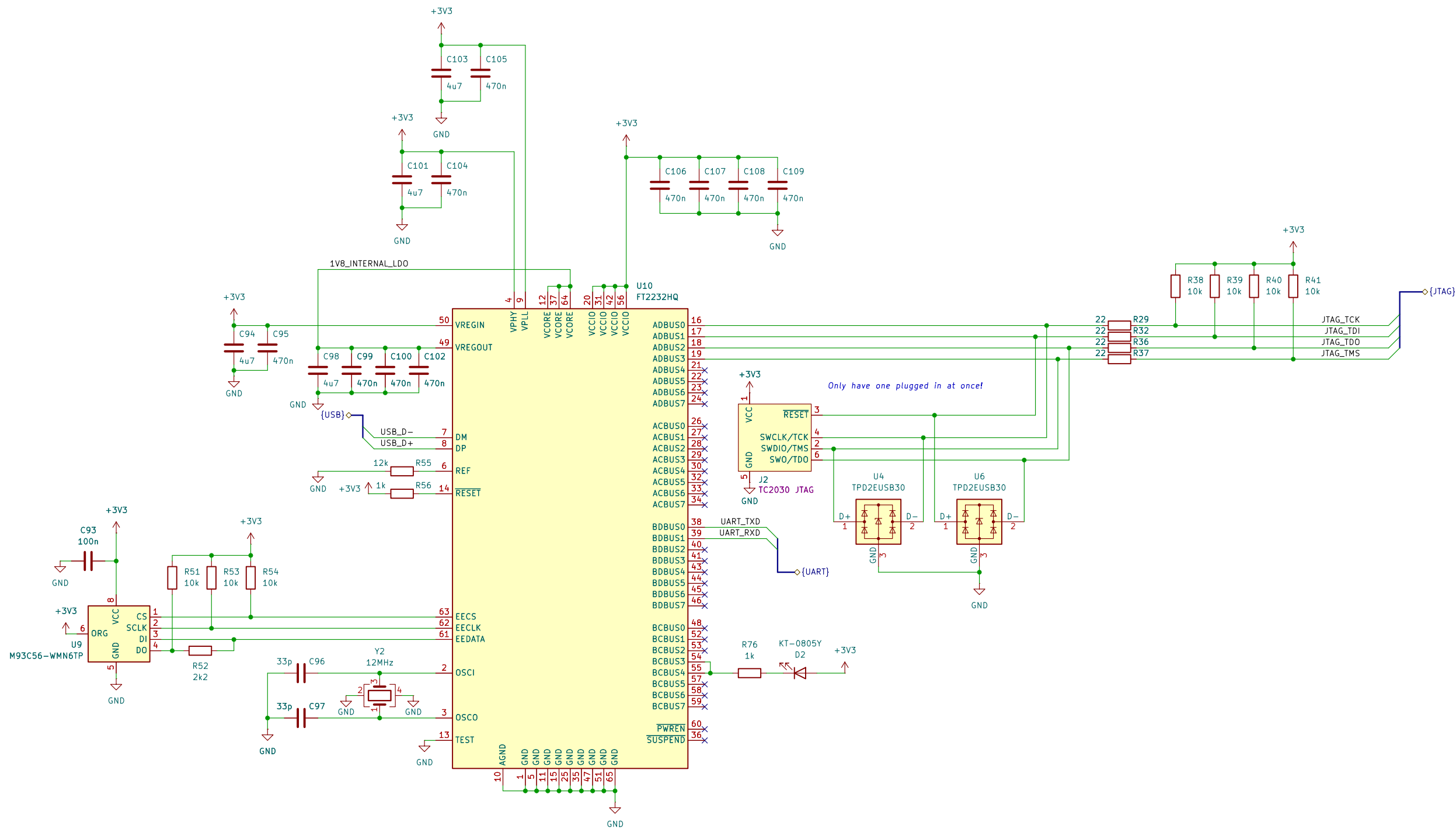


U1A
XC7A50T-FGG484

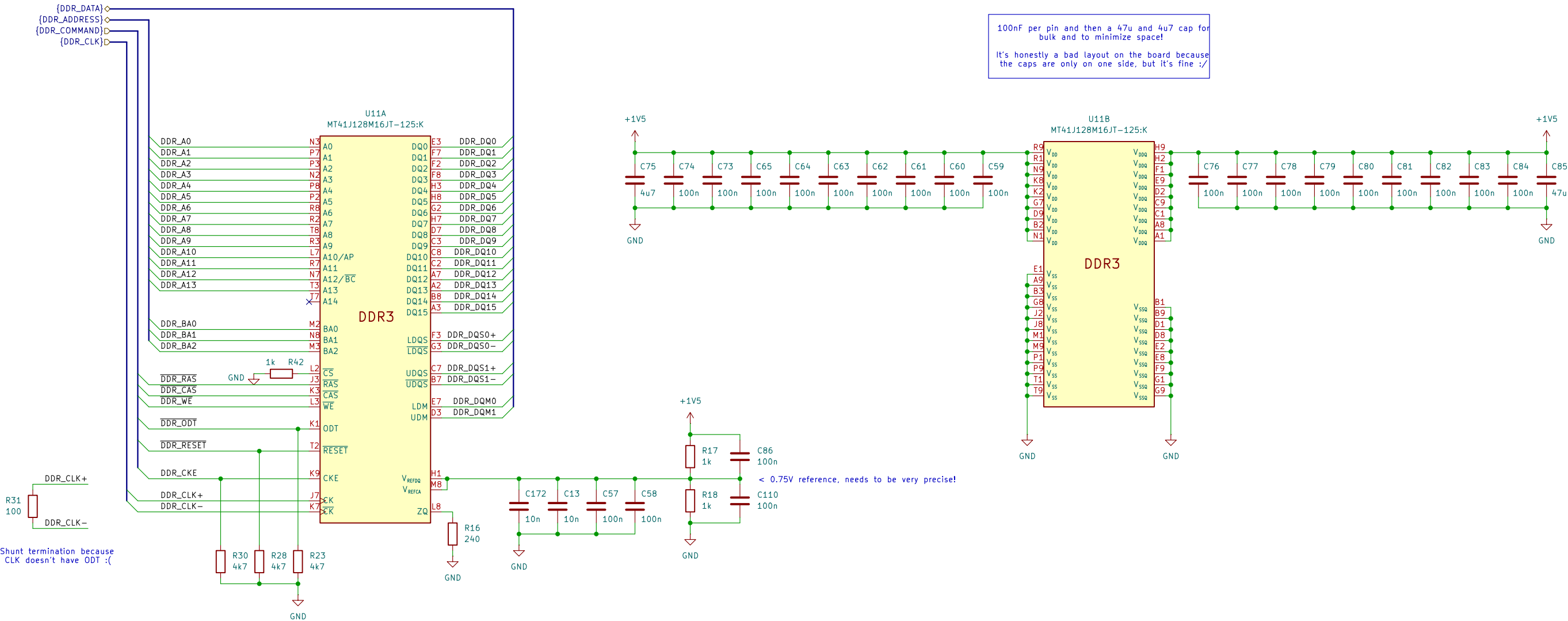


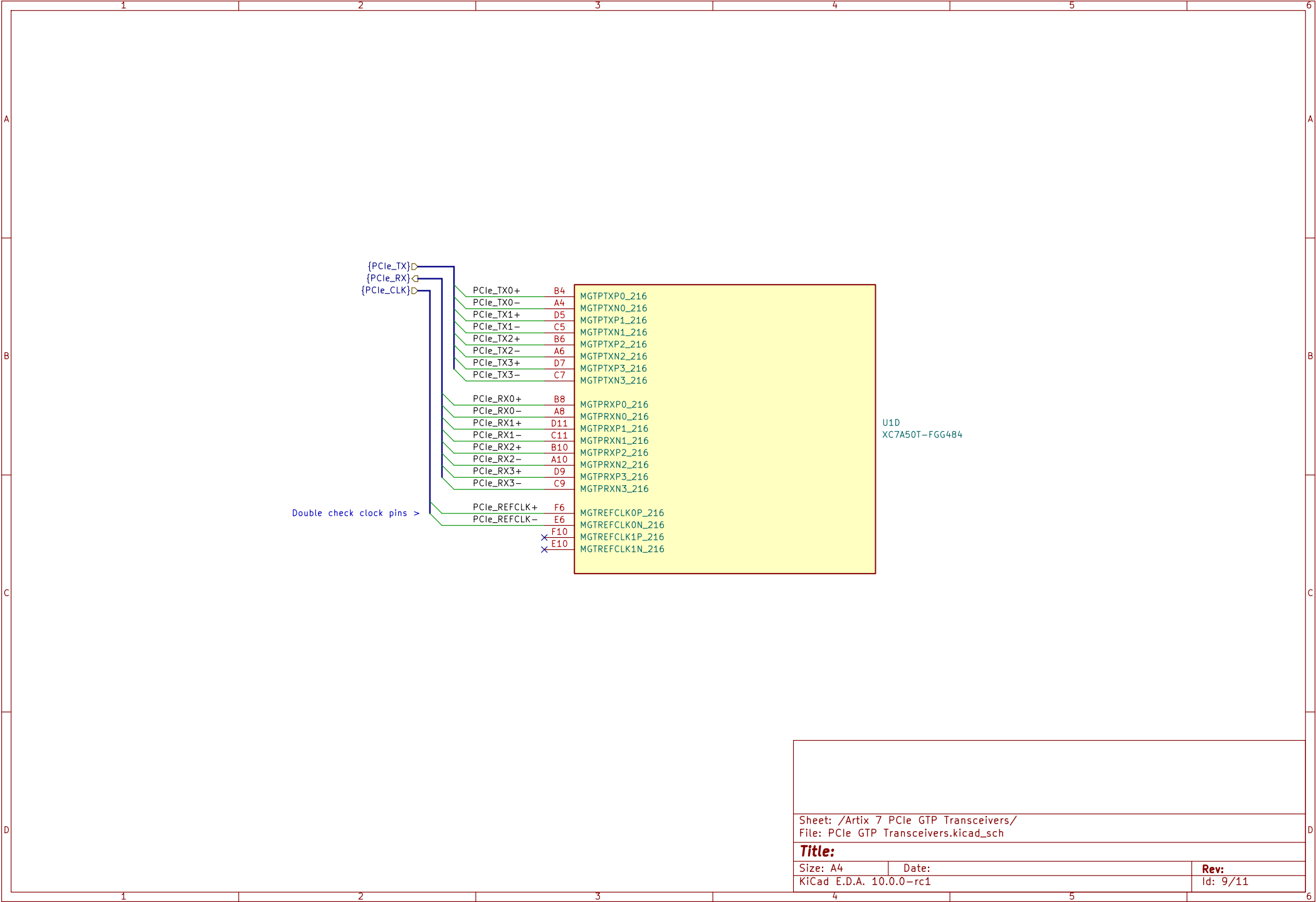
U1B
XC7A50T-FGG484





MT41J128M16JT-125:K
16 Bit data width
8 Bank machines
Normal ordering





Sheet: /Artix 7 PCIe GTP Transceivers/
File: PCIe GTP Transceivers.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.0-rc1

Rev:

Id: 9/11

